

Project 2
Due: 2/15/2018
ID Number:

Front-End Switches (100 points)

1) Design a switch that uses 3 major sampling techniques differential signaling, differential signaling + bottom-plate sampling and differential signaling+bootstrapping. The objective: large SFDR.

2) Rules of the game:

- Use transistors and capacitors. No verilog / amplifiers in this design.
- Generate the differential inputs using same setup as in project 1.
- Sample the voltage across capacitor for output and subtract with the differential side.
- Input frequency – ensure that it is within 0.1-1MHz and is a prime number frequency, 10 MHz clock.
- **In the bootstrapping, you should aim to achieve up to 16-bit dynamic range (thermal noise should be below the noise floor). Refer IEEE explore if need to. – Not in Phase 1.**

Use DFT function in Cadence calculator to show the frequency response and mark the SFDR of the signal cleanly.

3) Turn in:

- 1) schematics (legible so I can see the nodes that the skill code samples)
- 2) transients waveform of input, clocks and output samples
- 3) small bit of commentary

20 points: Final SFDR number and presented idea on circuit techniques

40 points: effective use of bootstrapping

10 points: effective use of differential sampling

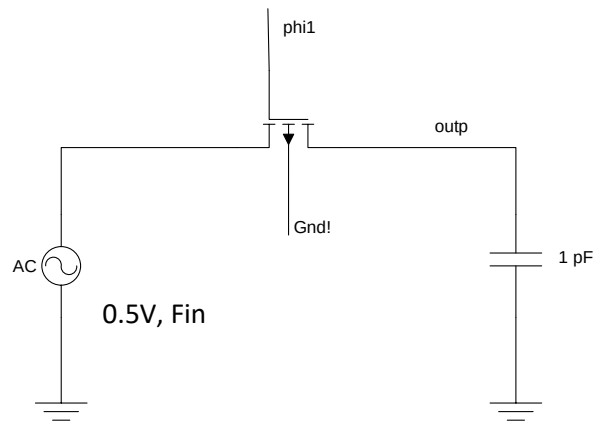
10 points: effective use of bottom-plate sampling (need to see clocks waveforms)

20 points: miscellaneous: clean FFT, code accuracy, proper inputs and data processing

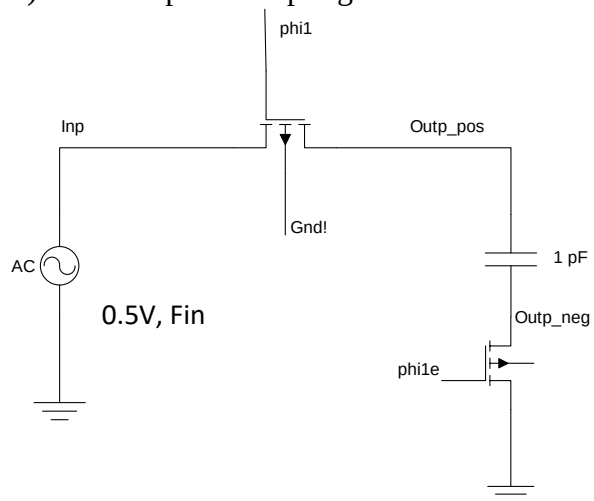
(Need to have both positive and negative signals for all the circuits)

Circuits:

- a) Differential Signaling:



b) Bottom-plate sampling:

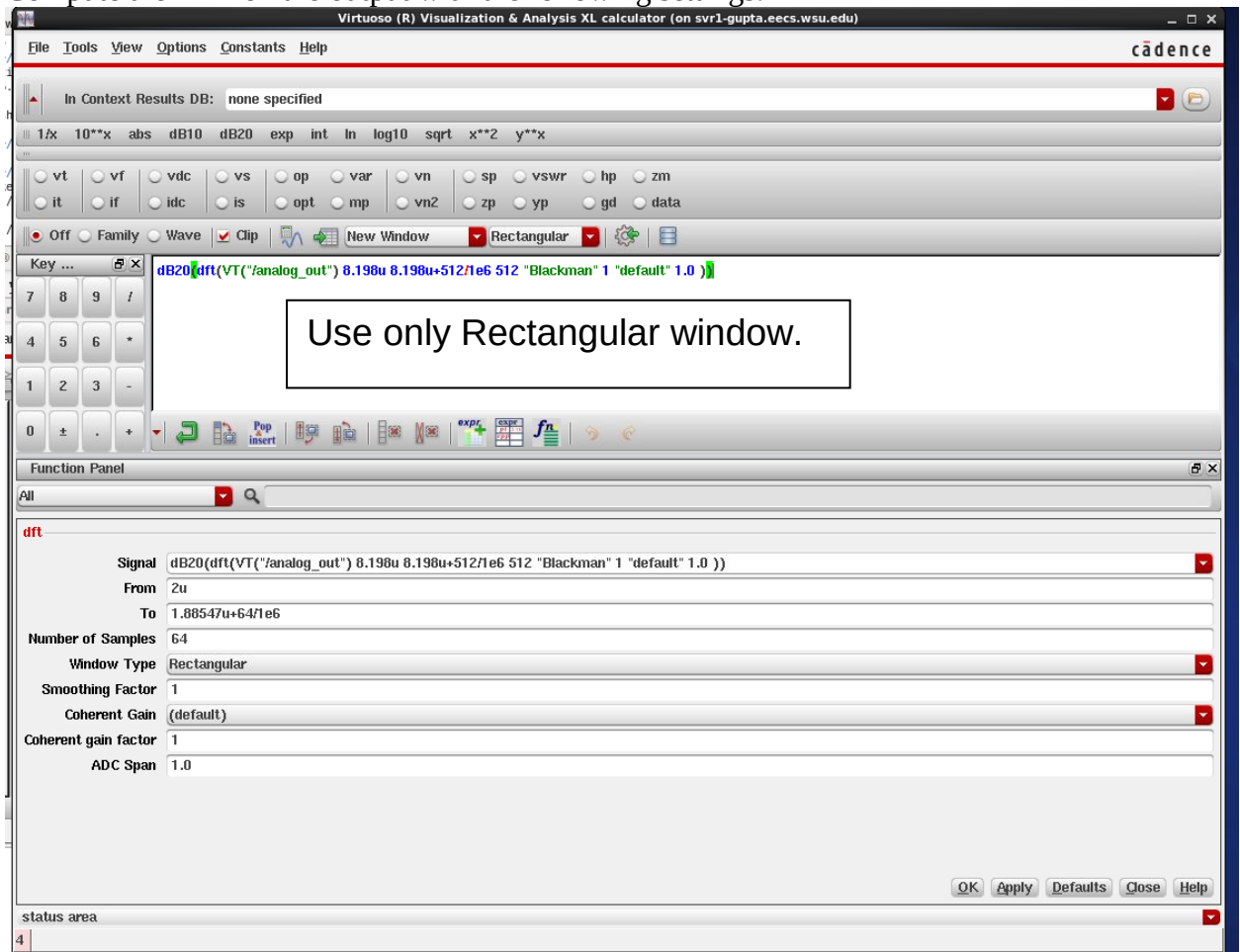


c) Bootstrapping w/ Bottom-plate sampling (together)

(Use Reference: Ref: A. M. Abo and P. R. Gray, "A 1.5-V, 10-bit, 14.3-MS/s CMOS pipeline ADC," IEEE Journal of Solid-State Circuits, vol. 34, issue 5, pp. 599-606, 1999.)

Tutorial: How to Plot SFDR:

- Select input frequency close to 0.1-1 MHz input but suitable for coherent sampling:
For eg:: $f_{in} = \text{prime\#} * F_{\text{sampling}} / 4096$ (e.g. prime #: 107)
You will be taking a 4096-pt FFT for example, with sampling frequency $F_{\text{samp}}=10$ MHz. You can select a prime number such that F_{in} is close to 0.1-1 MHz.
- Run the simulation
- Take the output across the capacitor and read it in the calculator.
- Compute the DFT on the output with the following settings:



Set t_{from} = time when the output on the capacitor is well settled,
 T_{to} = final time you will compute the data to = $t_{\text{start}} + 4096/f_s$
Number of samples = 4096
Window type = rectangular

The rest is set by default.

- After setting the DFT parameters, click OK and then click dB20.
- You will see a power spectral density plot. Select the main fundamental tone and then compute the SFDR against the highest harmonic and give the result in dBc (dB with respect to carrier).